

Analytic second derivatives of the Gielis boundary

pysie2d v0.6 — validation against Mie theory

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1 Scope

This note derives closed-form second derivatives $d^2 f/d\theta^2$, $d^2 g/d\theta^2$ of the Gielis super-formula boundary, replacing the first-order finite difference (`_der_real_3`) previously applied to the already-computed first derivatives. It is item 2 of the v0.6 architecture (`docs/design/v0.6-architecture.md`): analytic `ddf/ddg` is a prerequisite, alongside Kress–Martensen quadrature (item 1, not yet ported), for recovering spectral convergence on the logarithmic Green-function singularity. On its own — with the shipped Maradudin diagonal-patch quadrature still in place — it does not change the convergence *order*; that is confirmed, not assumed, in Section 6.

2 The Gielis super-formula

The boundary is parametrised by

$$\arg(\theta) = \frac{m\theta}{4}, \tag{1}$$

$$co(\theta) = \left| \frac{\cos(\arg)}{a} \right|^{n_2}, \quad se(\theta) = \left| \frac{\sin(\arg)}{b} \right|^{n_3}, \tag{2}$$

$$S(\theta) = co(\theta) + se(\theta), \tag{3}$$

$$r(\theta) = \text{rad} \cdot S(\theta)^{-1/n_1}, \tag{4}$$

$$f(\theta) = r(\theta) \sin \theta + x_0, \quad g(\theta) = r(\theta) \cos \theta + z_0. \tag{5}$$

3 First derivative (existing code, `_rderiv`)

Writing $\text{fact}_{n_2} = -n_2 m/4$, $\text{fact}_{n_3} = n_3 m/4$, and differentiating $|\cos(\arg)|^{n_2}$ using $d|\cos x|/dx = -\text{sign}(\cos x) \sin x$ gives the standard result

$$\frac{d co}{d\theta} = \text{fact}_{n_2} co \tan(\arg), \quad \frac{d se}{d\theta} = \text{fact}_{n_3} se \cot(\arg), \tag{6}$$

so that

$$S' = \text{fact}_{n_2} co \tan(\arg) + \text{fact}_{n_3} se \cot(\arg), \quad r' = \text{rad} \cdot p S^{p-1} S', \quad p := -\frac{1}{n_1}. \tag{7}$$

This is already implemented and unchanged by this work.

4 Second derivative (new, _rderiv2)

Differentiate S' a second time. For the cosine term,

$$\frac{d}{d\theta} \left[co \tan(\arg) \right] = \frac{d co}{d\theta} \tan(\arg) + co \sec^2(\arg) \frac{d \arg}{d\theta} \quad (8)$$

$$= \text{fact}_{n_2} co \tan^2(\arg) + co \sec^2(\arg) \cdot \frac{m}{4}, \quad (9)$$

using $d(\tan x)/dx = \sec^2 x$ and $d \arg/d\theta = m/4$. Multiplying by fact_{n_2} (the outer chain-rule factor already present in S'):

$$\left. \frac{d^2 S}{d\theta^2} \right|_{co\text{-term}} = co \left[\text{fact}_{n_2}^2 \tan^2(\arg) + \text{fact}_{n_2} \cdot \frac{m}{4} \sec^2(\arg) \right]. \quad (10)$$

Symmetrically, using $d(\cot x)/dx = -\csc^2 x$,

$$\left. \frac{d^2 S}{d\theta^2} \right|_{se\text{-term}} = se \left[\text{fact}_{n_3}^2 \cot^2(\arg) - \text{fact}_{n_3} \cdot \frac{m}{4} \csc^2(\arg) \right]. \quad (11)$$

Summing,

$$S'' = co \left[\text{fact}_{n_2}^2 \tan^2(\arg) + \text{fact}_{n_2} \frac{m}{4} \sec^2(\arg) \right] + se \left[\text{fact}_{n_3}^2 \cot^2(\arg) - \text{fact}_{n_3} \frac{m}{4} \csc^2(\arg) \right]. \quad (12)$$

With $r = \text{rad} \cdot S^p$, $p = -1/n_1$, the chain and product rule give

$$r'' = \text{rad} \cdot p(p-1) S^{p-2} (S')^2 + \text{rad} \cdot p S^{p-1} S''. \quad (13)$$

Finally, from $f = r \sin \theta$, $g = r \cos \theta$ (dropping the constant centre offset),

$$f'' = r'' \sin \theta + 2r' \cos \theta - r \sin \theta, \quad (14)$$

$$g'' = r'' \cos \theta - 2r' \sin \theta - r \cos \theta. \quad (15)$$

Validity. At $\cos(\arg) = 0$ the term $co \tan^2(\arg) \sim |\cos(\arg)|^{n_2-2}$: the removable limit is 0 for $n_2 > 2$, a finite nonzero constant at $n_2 = 2$, and a genuine divergence (a true corner) for $n_2 < 2$. The same holds for the se term at $\sin(\arg) = 0$ with n_3 . This is not a masking artefact of the implementation; it is where the boundary itself stops being C^2 . It matches the release's stated scope (`docs/conventions.md`): true corners are excluded, and every shape actually exercised in v0.6 so far has $n_2, n_3 \geq 2$.

5 Validation 1: closed form vs. an independent finite difference

The closed form (Eqs. 12–15) is checked against a 5-point, $O(h^4)$ centred finite difference applied directly to $f(\theta)$, $g(\theta)$ from the super-formula — an independent method, not agreement between two paths sharing the same assumptions. The step $h = 10^{-3}$ minimises the float64 truncation/round-off trade-off for this formula (truncation $\sim h^4$, round-off $\sim \epsilon/h^2$); the measured floor is $\sim 1.6 \times 10^{-8}$ relative.

This comparison is also encoded as `tests/test_geometry.py`, which compares f'' , g'' (not just r'') on the actual node grids the solver uses, at `rto1` = 10^{-6} .

Shape	m	(n_1, n_2, n_3)	max relative error, $d^2r/d\theta^2$
Circle	0	(2,2,2)	0 (trivial: $\text{fact}_{n_2} = \text{fact}_{n_3} = 0$)
Ellipse ($a=1.3, b=0.8$)	4	(2,2,2)	3.0×10^{-5}
Superellipse	4	(4,4,4)	3.7×10^{-5}
Superellipse	4	(8,8,8)	5.7×10^{-6}
Star (out of scope)	4	(1,1,1)	diverges at $\cos(\arg) = 0$

Table 1: Analytic r'' vs. a 4th-order finite difference on $r(\theta)$ directly. The $(n=1, 1, 1)$ row is the expected true-corner failure of Section 4, not a defect.

6 Validation 2: end-to-end against analytic Mie theory

The Mie-anchored regression tests (`test_efficiencies.py`, `test_mie_qnm.py`) all happen to exercise the boundary at $m = 0$, where $\text{fact}_{n_2} = \text{fact}_{n_3} = 0$ and the new formula collapses to $r'' \equiv 0$ trivially — the same answer the old finite difference gave. That is not a real test of Eq. 13. To validate the general formula end-to-end, the circle is instead built at $m = 4$, $n_1 = n_2 = n_3 = 2$: still an exact circle (`Geometry.is_circle` confirms $r = \text{rad}$ to 10^{-12} relative), but routed through the full non-trivial co , se , $\tan$, $\cot$ machinery.

n_n	Q_{ext} (BIE, analytic ddf/ddg)	Q_{ext} (Mie)	relative error
30	2.6060	2.6441	1.44×10^{-2}
60	2.6249	2.6441	7.27×10^{-3}
120	2.6345	2.6441	3.64×10^{-3}
240	2.6393	2.6441	1.82×10^{-3}
300	2.6403	2.6441	1.46×10^{-3}

Table 2: TE scattering efficiency, $\lambda = 600 \text{ nm}$, $n_{\text{core}} = 1.5$, $n_{\text{clad}} = 1.0$, $\text{rad} = 200 \text{ nm}$, circle built at $m=4$, $n_1=n_2=n_3=2$ so the analytic derivatives are exercised on their general branch.

The error roughly halves with each doubling of n_n — first-order convergence, exactly as expected: the shipped Maradudin diagonal-patch quadrature, not the derivative accuracy, still sets the rate. This confirms the analytic derivatives are *correct* and change nothing about the *order* on their own, matching the architecture doc’s framing that Kress–Martensen splitting (not yet ported) is the piece that raises the rate, with the analytic derivatives needed on top of it to reach machine precision.

7 Conclusion

The closed-form $d^2f/d\theta^2$, $d^2g/d\theta^2$ (`_rderiv2` in `src/pysie2d/geometry.py`) is validated two ways: against an independent finite-difference method directly on the defining formula, and end-to-end against analytic Mie theory on the non-trivial branch of the super-formula. Both confirm correctness. Neither is expected — nor found — to change the $O(h)$ convergence rate, which is set by the quadrature treatment of the log singularity (item 1, Kress–Martensen, still to be ported), not by derivative accuracy.